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Lightweight Real-Time Intrusion Detection for IoT Networks Using TSANet Deployed on Raspberry Pi with Live Traffic Monitoring

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Abstract

Intrusion Detection Systems (IDS) are essential for protecting IoT networks; however, many deep learning–based IDS solutions are too computationally intensive to be deployed on resource-constrained edge devices. This paper proposes TSANet, a lightweight Temporal–Spatial Attention Network for real-time intrusion detection on low-power hardware such as the Raspberry Pi 5. The proposed model integrates a 1D Convolutional Neural Network for spatial feature learning, an LSTM network for temporal sequence modeling, and attention mechanisms to highlight important traffic features and time steps. TSANet is trained and evaluated on the CICIDS2017 dataset and optimized using pruning and quantization-aware training to reduce model size and inference latency. Experimental results show that the optimized model achieves 98.3% offline accuracy and 96.8% real-time detection accuracy on Raspberry Pi, with an inference latency of 8–12 ms per flow, demonstrating its effectiveness for practical IoT edge-based intrusion detection.

Keywords: Intrusion Detection System; IoT Security; Edge Computing; Raspberry Pi; Deep Learning

1 Introduction

This work aligns with the United Nations Sustainable Development Goal 9 (SDG 9), which emphasizes building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation through secure and reliable digital systems. The rapid growth of Internet of Things (IoT) technologies has led to the large-scale deployment of connected devices in smart homes, healthcare systems, industrial automation, and smart cities. These devices continuously exchange data over networks and often operate in open or untrusted environments, making them attractive targets for cyberattacks such as denial-of-service (DoS), distributed denial-of-service (DDoS), malware injection, and data breaches. Ensuring the security of IoT networks has therefore become a critical challenge.

Intrusion Detection Systems (IDS) are widely used to monitor network traffic and identify malicious activities. Traditional IDS approaches rely on signature-based or rule-based techniques, which are effective for known attacks but fail to detect novel or evolving threats. In recent years, deep learning–based IDS solutions have demonstrated strong detection performance by learning complex patterns in network traffic. However, most existing deep learning models are computationally heavy and designed to run on powerful servers or cloud infrastructure.

Deploying such models directly on IoT edge devices is difficult due to limited processing power, memory, and energy availability. Transmitting raw network data to centralized servers also introduces latency and raises privacy concerns. These limitations highlight the need for lightweight, real-time intrusion detection systems that can operate directly on edge devices without relying on external computation.

To address these challenges, this work proposes TSANet, a lightweight Temporal–Spatial Attention Network designed for efficient deployment on resource-constrained hardware. The proposed system performs complete intrusion detection at the edge, including traffic analysis and real-time inference, and is validated through deployment on a Raspberry Pi 5 using both benchmark datasets and live network traffic. The results demonstrate that effective intrusion detection can be achieved at the IoT edge with low latency and high accuracy.

2 Related Work

Intrusion detection in Internet of Things (IoT) networks has gained significant research attention due to the rapid deployment of connected devices in smart cities, smart homes, and industrial environments. The integration of 5G networks and the use of resource-constrained edge devices further intensify security challenges. Existing research can be broadly categorized into traditional intrusion detection approaches, deep learning–based IDS, and lightweight or edge-deployable solutions for IoT environments, as summarized in Table 1.

2.1 Traditional and Early IDS Approaches

Early intrusion detection research primarily relied on conventional machine learning and rule-based techniques to identify malicious activities in network traffic. These approaches established the foundation for IDS development but struggle to scale effectively in heterogeneous and dynamic IoT environments. The importance of secure digital infrastructure has also been emphasized globally through initiatives such as the United Nations Sustainable Development Goal 9, which highlights resilient and secure infrastructure as a key requirement for sustainable development [1].

2.2 Deep Learning-Based Intrusion Detection Systems

To address the limitations of traditional methods, deep learning techniques have been widely adopted for intrusion detection in IoT networks. Idrissi et al. proposed a deep learning–based host IDS optimized for IoT environments, achieving improved detection accuracy through architectural optimization [2]. Similarly, Saba et al. introduced an anomaly-based IDS using deep learning models to enhance attack detection performance [3].

Hybrid architectures combining convolutional and recurrent models have also been explored. Gueriani et al. integrated CNN and LSTM networks to capture both spatial and temporal traffic characteristics [4], while Otoum and Nayak proposed a deep learning IDS framework emphasizing sequential learning for IoT security [5]. Survey studies further highlight the effectiveness of deep learning for IoT IDS while noting challenges related to model complexity and deployment feasibility [6, 7].

2.3 Lightweight and Edge-Based IDS Solutions

Recent research increasingly focuses on deploying intrusion detection systems directly on edge devices to reduce latency, bandwidth usage, and privacy risks. Kayode et al. proposed a Temporal–Spatial Attention Network (TSANet) for DoS attack detection, demonstrating the benefit of attention mechanisms in highlighting critical traffic features [8]. Racherla et al. developed a real-time deep IDS for IoT nodes, showing the feasibility of deep learning deployment in constrained environments [9].

Several studies specifically investigate Raspberry Pi–based IDS deployments. Dahiya et al. demonstrated a Raspberry Pi–based IDS for IoT networks [10], while Olakanmi and Olatunji proposed an optimized ensemble deep learning IDS achieving improved accuracy with reduced resource usage [11]. Adrian et al. designed an adaptive IDS for smart home security on Raspberry Pi platforms [12], and Gupta et al. explored deep learning–based IDS in 5G-enabled IoT networks, emphasizing low-latency detection requirements [13]. Despite these advances, many approaches lack fully integrated real-time traffic capture and inference on low-power hardware.

Table 1: Summary of IoT Intrusion Detection Systems

Author	**Methodology**	**Dataset**	**Feature Extraction**	**Results**
Idrissi et al. (2022) [2]	Optimized deep learning HIDS for IoT edge	IoT system logs	Host-level statistical features	Improved accuracy with reduced overhead on edge devices
Saba et al. (2022) [3]	Anomaly-based deep learning IDS	IoT traffic	Statistical flow features	High detection rate for IoT attacks
Gueriani et al. (2023) [4]	Hybrid CNN–LSTM IDS	Simulated IoT traffic	Spatial and temporal flow features	Enhanced detection accuracy over single models
Otoum and Nayak (2022) [5]	Deep learning–based IDS (DL-IDS)	IoT network traffic	Sequential traffic features	Effective detection of complex attacks
Kayode et al. (2025) [8]	Temporal–Spatial Attention Network (TSANet)	Network traffic traces	Attention-weighted flow features	High DoS detection accuracy using attention mechanisms
Racherla et al. (2024) [9]	Real-time deep IDS for IoT	IoT network traffic	Flow-based statistical features	Low-latency detection on IoT nodes
Dahiya et al. (2024) [10]	Raspberry Pi–based IDS	IoT traffic	Basic flow statistics	Feasible edge deployment with limited evaluation
Olakanmi and Olatunji (2024) [11]	Optimized ensemble deep learning IDS	IoT traffic	Reduced feature set	Improved accuracy with lower resource usage
Adrian et al. (2025) [12]	Adaptive ML-based IDS	Smart home traffic	Network flow features	Adaptive detection for smart home security
Gupta et al. (2022) [13]	Deep learning IDS for 5G-IoT	IoT over 5G networks	Traffic flow features	Low-latency detection in 5G-enabled IoT

Author	Methodology	Dataset	Feature Extraction	Results
Tsimenidis et al. (2022) [6]	Survey on DL-based IoT IDS	Multiple benchmarks	Comparative feature analysis	Identified challenges in DL IDS deployment
Morshedi et al. (2024) [7]	Deep learning IDS for IoT security	IoT datasets	Flow-level features	High accuracy but limited edge evaluation
Shone et al. (2018) [14]	Deep autoencoder-based IDS	NSL-KDD	Learned latent representations	Improved detection of unseen attacks

2.4 Research Gap

Although numerous intrusion detection systems have been proposed for IoT networks, most existing solutions rely on computationally intensive deep learning models designed for cloud or server-based environments, making them unsuitable for real-time deployment on resource-constrained edge devices. Many studies report high detection accuracy only under offline or simulated conditions and do not evaluate latency, resource usage, or robustness under live network traffic. In addition, several approaches focus on either spatial or temporal feature learning independently, without effectively combining both in a lightweight architecture. While techniques such as model compression and federated learning have been explored.

3 Proposed Methodology

This section presents the proposed lightweight intrusion detection framework based on the Temporal–Spatial Attention Network (TSANet). The key contribution of this work is the design and real-time deployment of an attention-driven deep learning IDS that operates entirely on a Raspberry Pi edge device using live network traffic. Unlike conventional IDS pipelines that rely on centralized processing, the proposed system performs end-to-end traffic capture, feature extraction, inference, and alert generation entirely at the edge.

3.1 System Overview

The overall workflow of the proposed system is illustrated in Figure 3. Network packets are captured in real time from live traffic, converted into bidirectional flows, processed into numerical features, and classified using the trained TSANet model deployed on a Raspberry Pi device. Based on the prediction confidence, alerts are generated locally without any dependency on cloud servers.

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3.2 Data Collection and Feature Engineering

Network traffic data is collected in two phases. During offline training, the CICIDS2017 benchmark dataset is used due to its realistic traffic patterns and diverse attack classes. For real-time evaluation, live traffic is captured directly on the Raspberry Pi configured as a Wi-Fi access point.

Each network flow is represented using 32 carefully selected bidirectional flow features, including packet statistics, flow duration, byte counts, and inter-arrival times. These features are chosen to balance detection effectiveness and computational efficiency, ensuring suitability for edge deployment. Feature scaling is performed using Min–Max normalization to maintain numerical stability during inference.

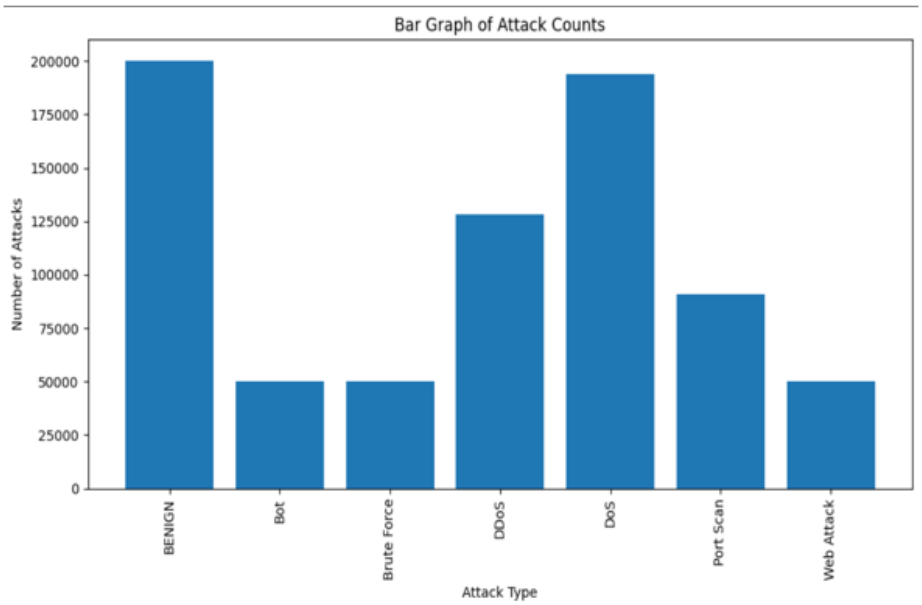

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Figure 1: Distribution of benign and attack samples in the CICIDS-2017 dataset

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Figure 1 illustrates the overall data distribution and serves as the basis for the proposed TSANet-based intrusion detection pipeline. The normalized flow features extracted from the CICIDS-2017 dataset and real-time traffic (as described in the previous subsection) are fed into the TSANet architecture for temporal feature learning and attack classification.

3.3 TSANet Architecture

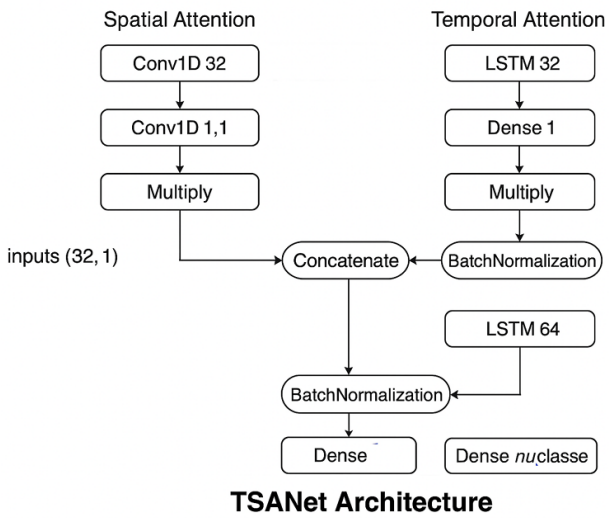

Figure 2: TSANet architecture illustrating spatial and temporal attention mechanisms

The core of the proposed system is the TSANet model, whose architecture is shown in Figure 2. TSANet combines convolutional, recurrent, and attention-based components to effectively model both spatial and temporal characteristics of network traffic.

The spatial attention branch applies a one-dimensional convolutional layer with 32 filters to capture local feature interactions across the input feature vector. A secondary 1×1 convolution generates attention weights, which are applied through element-wise multiplication to emphasize informative features while suppressing irrelevant ones.

In parallel, the temporal attention branch employs an LSTM layer with 32 hidden units to model sequential dependencies across time. A dense layer followed by a sigmoid activation produces temporal attention weights, allowing the model to focus on critical time steps within each flow. Batch normalization is applied to stabilize training and improve convergence.

The outputs of both attention branches are concatenated and passed through an additional LSTM layer with 64 units to jointly learn spatio-temporal representations. Finally, fully connected layers map the learned features to the output space, producing class probabilities for benign and attack traffic.

3.4 Model Training and Optimization

TSANet is trained using supervised learning with categorical cross-entropy loss. Given an input feature sequence $X = \{x_1, x_2, \ldots, x_T\}$ and the corresponding

ground-truth label y, the optimization objective is defined in Eq. (1).

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{1}$$

where $\hat{y}_c$ denotes the predicted probability for class c.

To enable efficient edge deployment, the trained model undergoes pruning and quantization-aware training, reducing model size and inference latency without significant loss in accuracy. The optimized model is exported using TorchScript, ensuring compatibility with low-power hardware environments.

3.5 Edge Deployment on Raspberry Pi

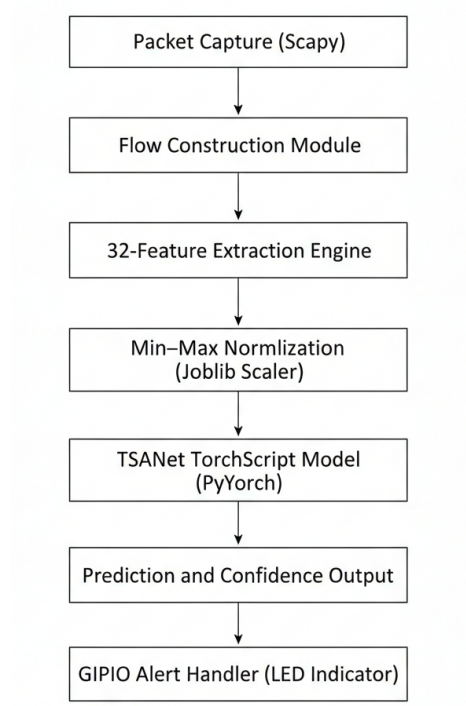

Figure 3: End-to-end system workflow for real-time traffic capture, feature extraction, and edge inference

The optimized TSANet model is deployed on a Raspberry Pi 5, configured to operate as a live network gateway. Packet capture is performed using Scapy, followed by on-device flow reconstruction and feature extraction. Inference is executed locally using the TorchScript model, ensuring low-latency detection.

Figure 3 illustrates the complete runtime pipeline, including packet capture, feature processing, model inference, and alert generation. Upon detecting ma-

licious activity, the system outputs prediction confidence and triggers a GPIO-based LED indicator for immediate physical alerting. This fully autonomous design eliminates reliance on external servers and preserves data privacy.

Overall, the proposed methodology demonstrates that effective intrusion detection can be achieved directly at the IoT edge by integrating attention-driven deep learning with lightweight deployment strategies.

4 Results and Discussion

This section presents the experimental results of the proposed TSANet-based intrusion detection system, evaluated using both benchmark datasets and real-time deployment on a Raspberry Pi edge device.

4.1 Offline Evaluation on CICIDS2017

The proposed TSANet model was first evaluated offline using the CICIDS2017 dataset. Performance was measured using standard classification metrics including accuracy, precision, recall, and F1-score.

Table 2 compares TSANet with commonly used baseline models. The proposed model achieves the highest overall accuracy of 98.3%, outperforming traditional machine learning and deep learning baselines. This improvement is attributed to the combined spatial–temporal attention mechanism, which enables selective focus on important features and time steps.

Table 2: Comparison of TSANet with Baseline Models

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**
Random Forest	94.2	93.7	92.5	92.8
CNN	95.6	95.1	94.8	94.9
LSTM	96.2	96.0	95.7	95.8
CNN–LSTM	97.0	96.9	96.4	96.6
Proposed TSANet	**98.3**	**98.6**	**97.9**	**98.1**

4.2 Effect of Model Pruning

To optimize TSANet for edge deployment, structured pruning was applied. Figure 4 shows the relationship between pruning rate and classification accuracy. The results indicate that accuracy remains stable up to moderate pruning levels, demonstrating that TSANet retains robustness even after model compression.

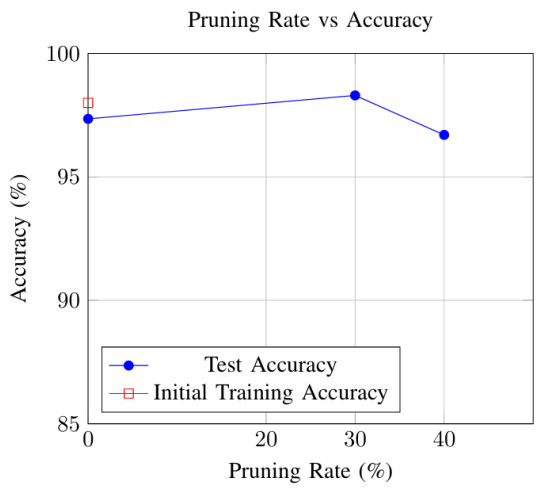

Figure 4: Impact of pruning rate on classification accuracy

4.3 Confusion Matrix Analysis

Table 3 presents the confusion matrix of the optimized TSANet model on CICIDS2017. High detection rates are observed for volumetric attacks such as DoS and DDoS. Slight misclassification is noted for sparse attack classes such as Heartbleed, mainly due to limited flow-level evidence.

Table 3: Confusion Matrix of Optimized TSANet on CICIDS2017 (%)

Actual / Predicted	Benign	Bot	BF	DDoS	DoS	Heartbleed	Infiltration
Benign	98.1	0.2	0.3	0.1	0.8	0.0	0.5
Bot	0.4	97.6	1.2	0.0	0.3	0.0	0.5
Brute Force	0.6	0.8	96.4	0.0	1.7	0.0	0.5
DDoS	0.0	0.0	0.0	98.9	1.1	0.0	0.0
DoS	0.4	0.1	0.3	1.5	97.3	0.1	0.3
Heartbleed	0.0	0.0	0.0	0.0	2.5	95.8	1.7
Infiltration	1.3	0.4	0.0	0.0	0.7	0.2	97.4

4.4 Real-Time Deployment and Live Traffic Evaluation

To overcome the limitations of Mininet-based experiments, the optimized TSANet model was deployed on a Raspberry Pi 5 configured as a standalone wireless access point. Multiple laptops and mobile devices were connected to the access point, generating genuine Wi-Fi traffic through web browsing, messaging,

YouTube streaming, system updates, and background processes. Since all packets passed directly through the Raspberry Pi, complete bidirectional flows were captured, enabling accurate feature extraction and real-time prediction.

During live operation, each network flow was identified by its source IP address and port communicating with a destination IP address and port. For every such IP-to-IP flow observed at the access point, the system reconstructed flow-level features and passed them to the trained TSANet model, which classified the flow as benign or malicious and produced a corresponding confidence score.

A representative excerpt of real-time alerts generated during live traffic monitoring is shown below. The output demonstrates how traffic flowing from one IP address to another is analyzed and labeled by the system in real time.

```
===== ALERT =====
Flow: ('163.70.144.8', 443) → ('192.168.50.100', 45190)
Prediction: BENIGN (conf=0.5827)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: Heartbleed (conf=0.7354)
=================

===== ALERT =====
Flow: ('192.168.50.140', 52550) → ('34.107.221.82', 80)
Prediction: BENIGN (conf=0.5482)
=================

===== ALERT =====
Flow: ('192.168.50.140', 8080) → ('192.168.50.33', 41076)
Prediction: DDoS (conf=0.7714)
=================
```

In this output, each alert corresponds to a single network flow observed at the access point, where traffic is transmitted from a source IP address to a destination IP address. The model predicts whether the observed flow is normal (benign) or belongs to a specific attack class such as Heartbleed or DDoS, along with a confidence value indicating the model's certainty. Benign traffic generated by normal user activity is correctly identified, while attack traffic triggers high-confidence alerts.

Across several hours of continuous monitoring, the system achieved an average real-time detection accuracy of 96.8%. Detection latency remained within 8–12 ms per flow, and CPU and memory utilization on the Raspberry Pi 5 remained stable throughout operation. These results confirm that deploying the Raspberry Pi as a real access point provides the packet timing accuracy and flow completeness required for reliable intrusion detection, validating the practicality of the proposed TSANet-based IDS for real-world IoT environments.

5 Conclusion and Future Work

This work presented TSANet, a lightweight Temporal–Spatial Attention Network for real-time intrusion detection in IoT environments, designed for deployment on resource-constrained edge devices. By integrating spatial and temporal attention mechanisms with CNN and LSTM layers, the proposed system effectively captures discriminative traffic features while maintaining low computational overhead. Experimental evaluation on the CICIDS2017 dataset demonstrated superior detection performance compared to baseline models, achieving 98.3% offline accuracy. Real-time deployment on a Raspberry Pi configured as a Wi-Fi access point validated the practicality of the approach, with low inference latency (8–12 ms per flow) and stable resource usage during live traffic monitoring. The system successfully detected both benign and malicious activities without reliance on cloud infrastructure. As future work, the model can be extended to support additional attack classes and adaptive learning for evolving threats, along with evaluation in large-scale distributed IoT environments.

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